

Deep Learning

00 Introduction and Course logistics

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Course Logistics

- R slot
- Tuesday 2:30 - 4 PM
- Friday 4 - 5:30 PM
- Tentative Classroom : LHC-13

Course Logistics

- Teams Classroom (Deep Learning (CS5480/AI5100/AI2100): **05yhw6g**



Deep Learning 2025 (CS5480/AI5100/AI2100)

Channels

Settings

+5



Choose who can use team, channel, and group @mentions



Team code

Share this code so people can join the team directly - you won't get join requests

05yhw6g

Tentative Evaluation

- Assignments - 20%
 - 2-3 assignments, submission deadline on first Saturdays
- Written exams - 30%
 - 2-3 exams, second Saturdays every month
- Kaggle Project - 20%
 - Ranking (public+private), implementation (code), report
 - Group project (2-3)
 - Submission deadlines on penultimate week of April
- Endsem (written) - 30%
 - Last week of April
- No makeup exams! Penalties for missing submission deadlines !

Deep Learning Contents

- Introduction to **artificial neuron models** and neural network architectures
- Basic neuron models:
 - McCulloch–Pitts neuron
 - Perceptron
- Core neural network architectures:
 - Multi-Layer Perceptron (MLP)
 - Convolutional Neural Networks (CNNs)
 - Recurrent Neural Networks (RNN, LSTM, GRU)
- Training methods:
 - Gradient Descent
 - Backpropagation
- Advanced architectures:
 - Encoder–Decoder models
 - Attention mechanisms and variants
 - Self-Attention and Transformers
- Generative AI models:
 - Variational Autoencoders (VAEs)
 - GANs
 - GPT, BERT

Prerequisites

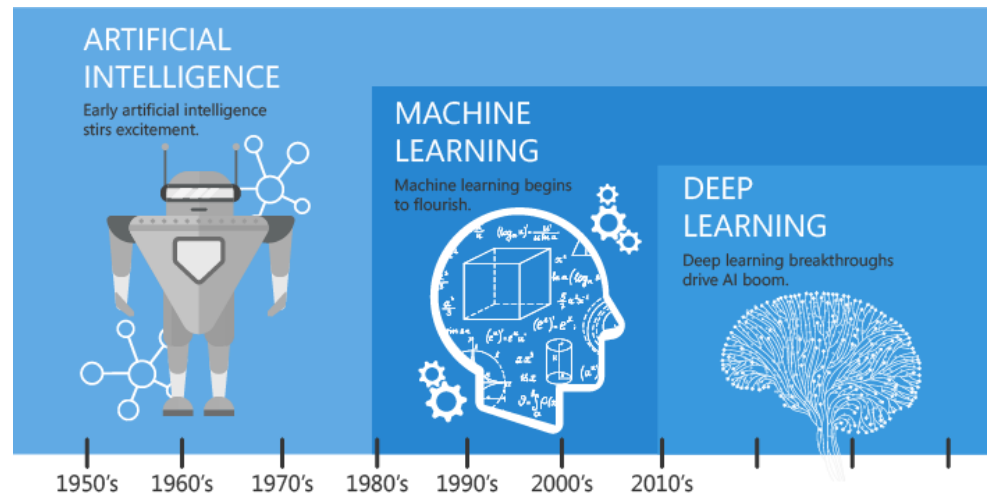
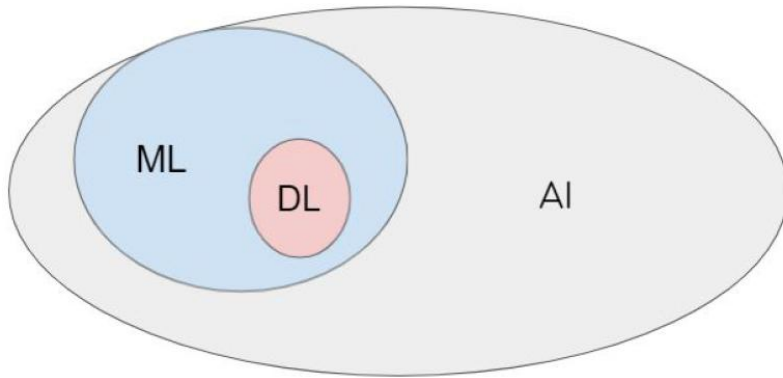
- Programming in Python
- A course on Machine Learning (CS3390, CS5590, AI2000, AI5000)

Textbooks and References

- Deep Learning textbook by Ian Goodfellow et al.
- Deep Learning Foundations and Concepts by C Bishop & H Bishop
- Michael Nielsen's text book on NN & DL
- PyTorch - <https://pytorch.org/>
- Several online resources and courses

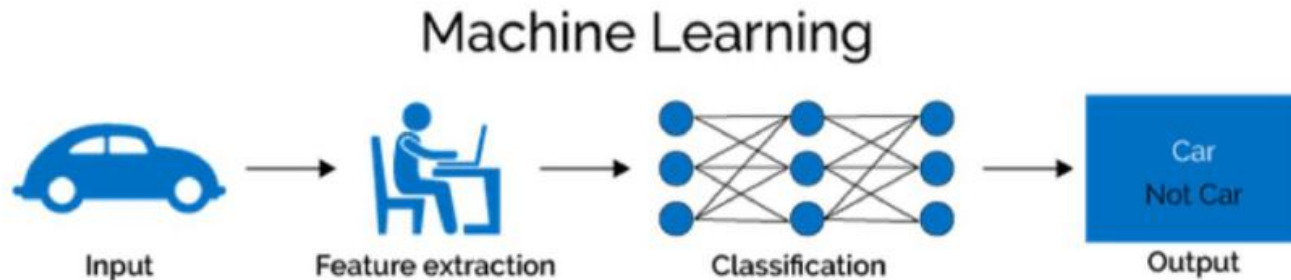
What is DL?

- Subset of ML that is essentially Artificial Neural Networks with more layers
- Crude attempt to imitate the human brain in learning



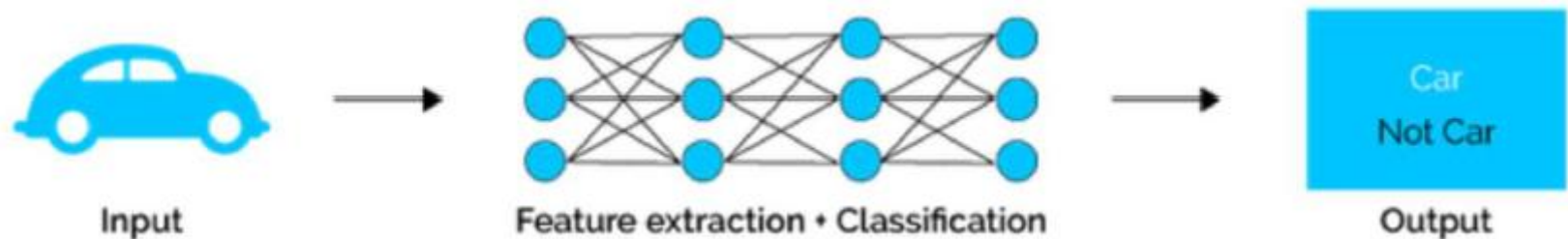
Classical ML vs. DL

- Classical ML: Handcrafted features + learnable model
- Need strong domain expertise



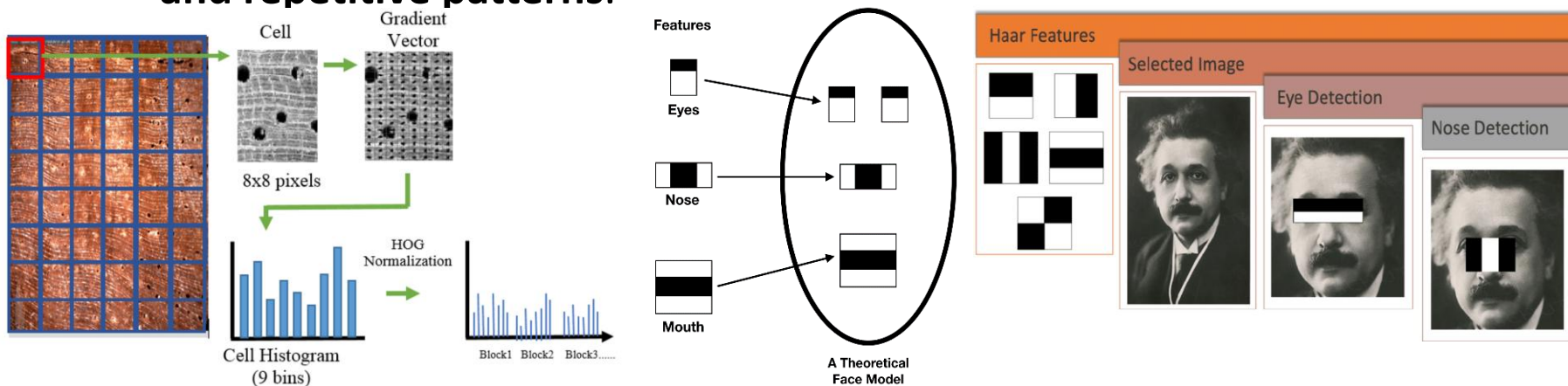
Classical ML vs. DL

- Deep Learning: Deep stack of parameterized processing
- End-to-End learning



Problem setting : Face detection

- Classical Machine Learning Pipeline
 - Hand-crafted Feature Extraction
 - HOG (Histogram of Oriented Gradients) : **local distributions of edge orientations**, capturing shape and appearance and robust to illumination changes.
 - **Haar-like features** : Differences between pixel intensities in rectangular regions
 - **Gabor features** are **handcrafted, frequency- and orientation-selective features** used in computer vision to capture **localized texture, edges, and repetitive patterns**.



Problem setting : Face detection

- Classical Machine Learning Pipeline

- Hand-crafted Feature Extraction

- $\phi(\cdot)$: manually designed feature extractor
 - $\mathbf{x}$: feature vector

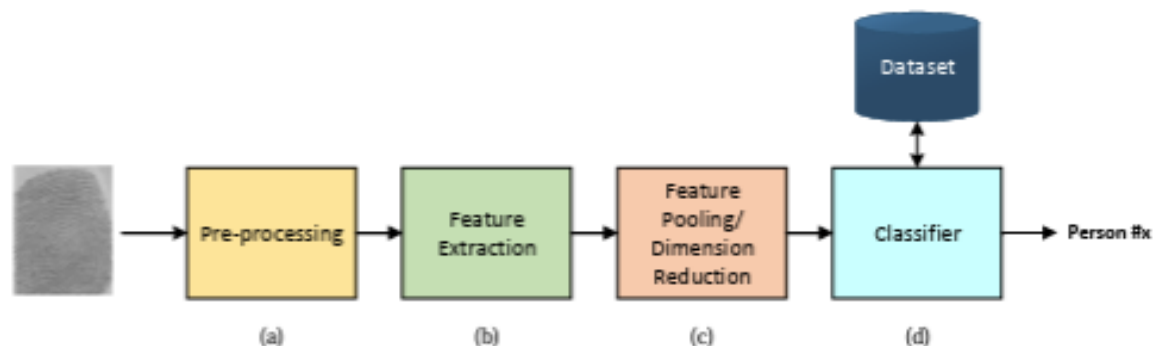
$$\mathbf{x} = \phi(\mathbf{I})$$

- Classifier Learning : learns **decision boundaries**, not representations.

- Support Vector Machine (SVM)
 - AdaBoost
 - Logistic Regression

$$\hat{y} = f(\mathbf{x}; \theta)$$

Gabor filter



Problem : Face Detection

Limitations

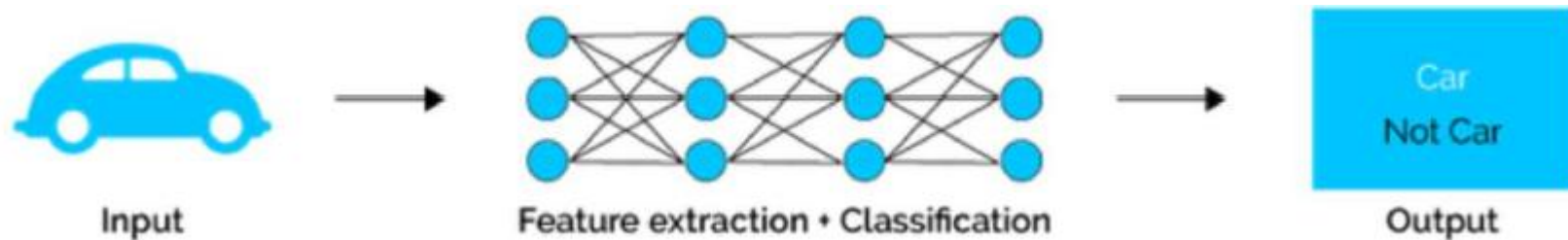
- Requires **domain expertise**
- Fragile to:
 - Pose changes
 - Lighting variation
 - Occlusion
 - Different ethnicities and ages
- Different handcrafted features for different problems
- **Classical machine learning asks:** *“What features should I look for?”*
Deep learning answers: *“Let me learn the features myself.”*

Problem : Face detection

Deep Learning

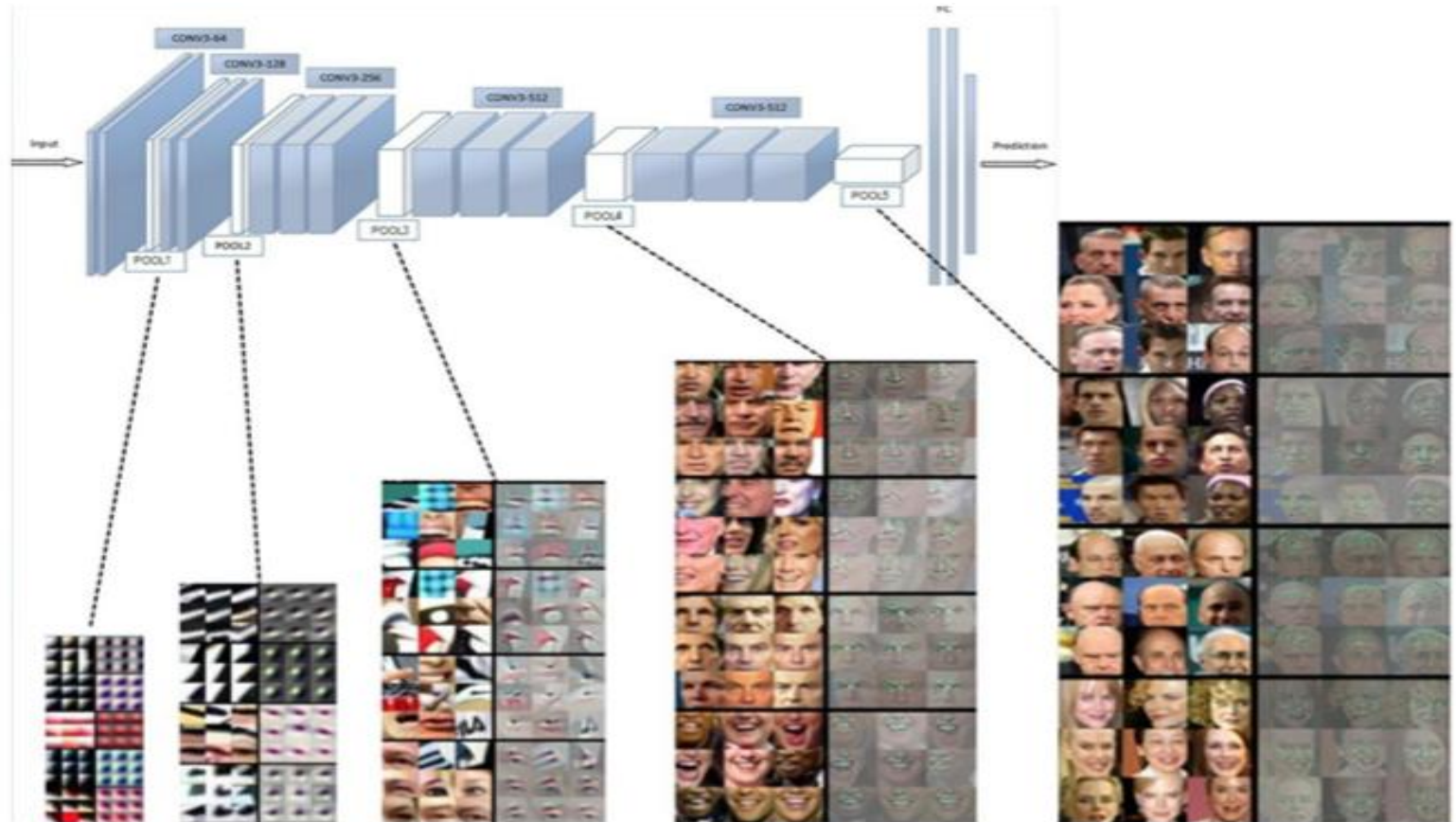
The model learns both features and the classifier end-to-end from data.

- Learn multiple levels of representations that correspond to different levels of abstraction, which is better suited to uncover underlying patterns of the data
- levels form a hierarchy of concepts, showing strong invariance to the face pose, lighting, and expression changes



Problem : Face detection

Deep Learning



Natural Language Processing

- Classical Machine Learning
 - Consider a one hot vector representation of documents (variants such as term frequency, tf-idf)
 - Apply ML models such as support vector machine

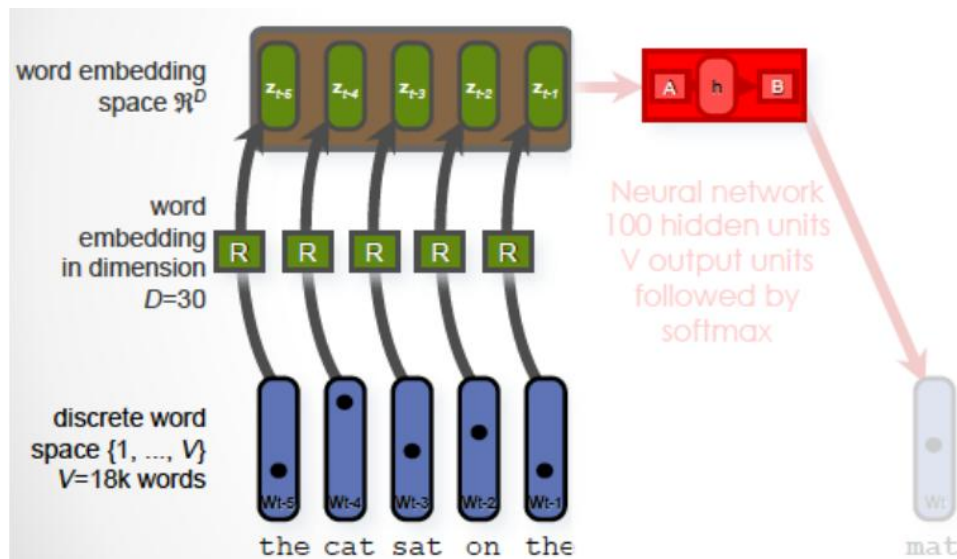
	Dog	Cat	Python	...
Document 1	1	1	0	...
Document 2	1	0	0	...
Document 3	0	0	1	...

Does not capture semantic similarity

Dog = [1,0,0,...,...] Cat = [0,1,0,...,0] Python = [0,0,1,...,0]

Neural networks for NLP

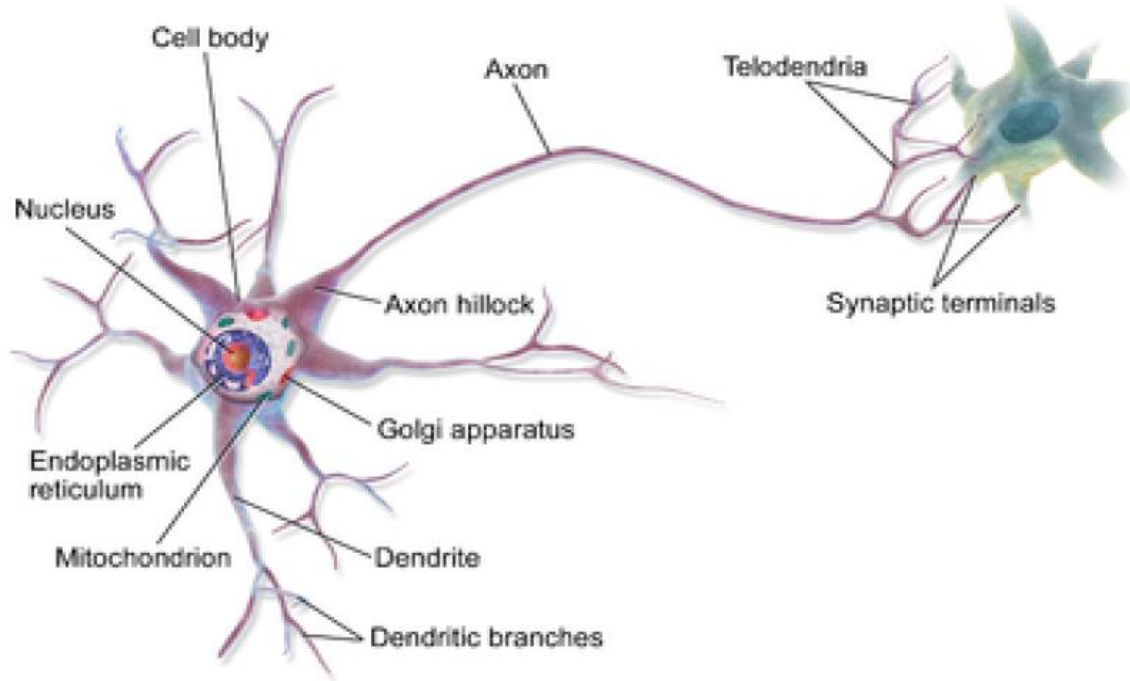
- Bengio et al. (2003) learnt embeddings of words (representation in a lower dimensional space) using a neural language model, where they use a neural network to predict next word in a sequence using past words
- Allows us to learn word embeddings with semantic interpretation and improves generalization performance



Word	w	$C(w)$
" the "	1	[0.6762, -0.9607, 0.3626, -0.2410, 0.6636]
" a "	2	[0.6859, -0.9266, 0.3777, -0.2140, 0.6711]
" have "	3	[0.1656, -0.1530, 0.0310, -0.3321, -0.1342]
" be "	4	[0.1760, -0.1340, 0.0702, -0.2981, -0.1111]
" cat "	5	[0.5896, 0.9137, 0.0452, 0.7603, -0.6541]
" dog "	6	[0.5965, 0.9143, 0.0899, 0.7702, -0.6392]
" car "	7	[-0.0069, 0.7995, 0.6433, 0.2898, 0.6359]

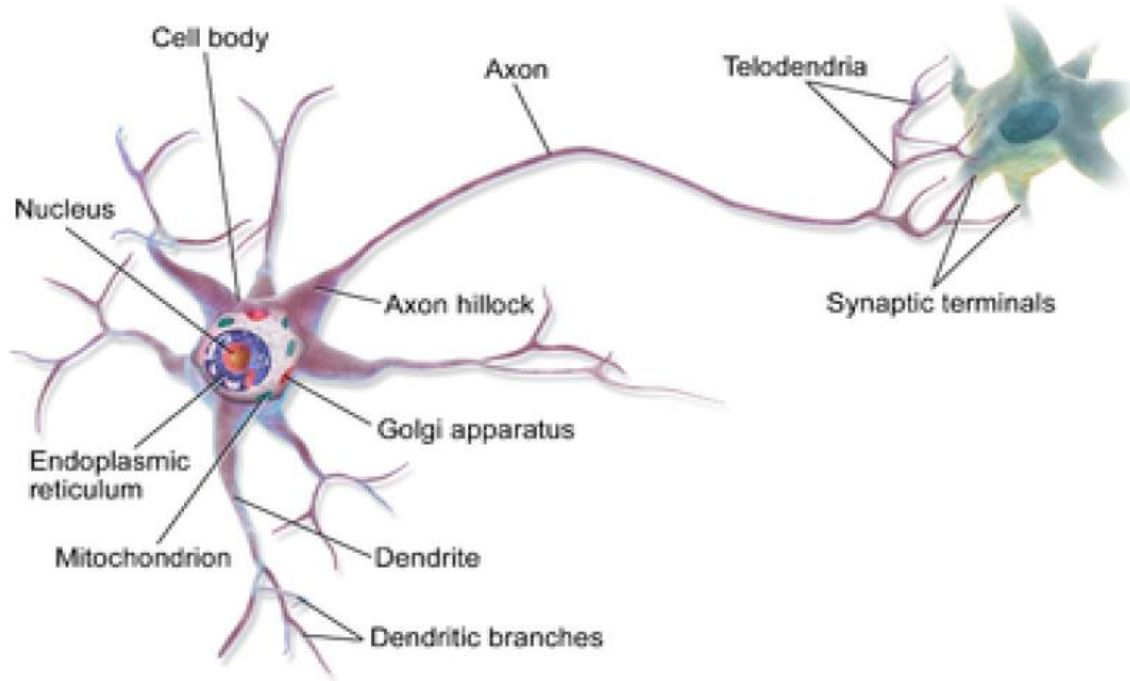
The Biological Neuron

- Approximately 86 billion neurons and an estimated 100 trillion synaptic connections.



The Biological Neuron

- Approximately 86 billion neurons and an estimated 100 trillion synaptic connections.
- GPT-4 has an estimated 1.7 trillion to 1.8 trillion parameters



The future of AI is not a bigger brain—it is a better organized one

History of Neural Networks

1. [McCulloch Pitts neuron](#) (1943) - Threshold Logic Unit

In the early 1940s, neuroscience and logic were largely separate disciplines. McCulloch (a neuroscientist) and Pitts (a logician) unified them by asking:

“Can the brain be understood as a logical computing machine?”

They proposed the **first mathematical model of a neuron**, now called the **McCulloch–Pitts (MP) neuron**

$$y = \begin{cases} 1 & \text{if } \sum_i w_i x_i \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

2. [Donald Hebb](#) (1949) - Hebbian Learning Principle

*If neurons compute, **how do they learn?***

Earliest principles of learning in neuroscience and artificial neural networks.

“Neurons that fire together, wire together.”

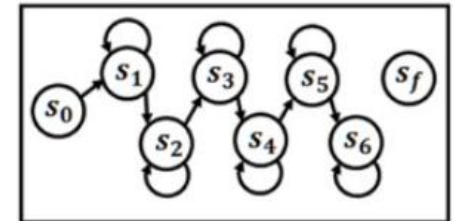
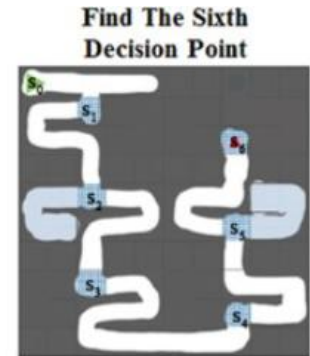
strength of a synapse between two neurons increases if the presynaptic neuron repeatedly contributes to the firing of the postsynaptic neuron.

=> Weight connecting a neuron to output increases if they exhibit correlated activity

$$\Delta w_{ij} \propto x_i \cdot y_j$$

History of Neural Networks

1. Marvin Minsky (1951) - created the first ANN
 1. Built neural network hardware system containing 40 neurons
 2. Followed reinforcement learning
 3. Simulated rat navigation
 4. Stochastic Neural Analog Reinforcement Calculator (SNARC) - analog machine built from approximately 3,000 vacuum tubes, a couple of hundred relays, motors, clutches, and potentiometers.



1. [Frank Rosenblatt](#) (1958) – Introduced perceptron – first trainable classifier to classify 20X20 images
 - Supervised learning
 - Error-driven updates
 - First **learning algorithm** with convergence guarantees (for linearly separable data)
 - Introduced **pattern recognition**

$$y = \text{sign} \left(\sum_i w_i x_i \right)$$

$$w \leftarrow w + \eta(y - \hat{y})x$$

SNARC demonstrated that networks can adapt; the perceptron demonstrated that networks can be trained.

AI over optimism and Winter

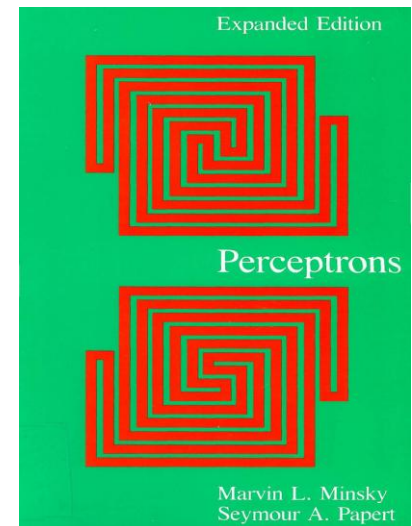
“Machines will match human intelligence in 20 years”

“Vision and language will be solved problems”

Perceptron Limitation and the Minsky–Papert Critique

Single-layer perceptrons **cannot represent non-linearly separable functions**

- In 1969, **Perceptrons** by **Marvin Minsky** and **Seymour Papert**:
- **Mathematically proved** fundamental limitations of perceptrons
- Showed many vision-relevant tasks were impossible for single-layer models
- Emphasized lack of learning algorithms for multilayer networks



- **Funding agencies misinterpreted the critique as “Neural networks don’t work”**
- Research focus shifted almost entirely to **symbolic AI**
- Neural network funding nearly vanished

AI Winter – 1970s to 80s

Backpropagation

How do you train hidden layers in a multilayer neural network?

- Revived interest in AI
- Provided an approach to find how each hidden weight contributed to the final error.
- Use the **chain rule of calculus** to propagate error gradients **backward** through the network.
- It Unified Neural Networks with Optimization
- **Paul Werbos** introduced backpropagation **first**:
 - PhD thesis (1974, Harvard) but not published in journal
 - Explicit use of the chain rule
 - Proposed training multilayer networks
 - Community bias against neural networks post-AI winter
- Backpropagation became influential after the 1986 paper by **David Rumelhart, Geoffrey Hinton and Ronald Williams**
 - Clear algorithmic formulation
 - Compelling experimental results
 - Demonstrated learning of internal representations

Paul Werbos (1974) invented backpropagation. Rumelhart, Hinton, and Williams (1986) demonstrated it convincingly and made it central to AI.

If you are working in AI, publish it and just don't leave it as your thesis work !

Backpropagation

How do you train hidden layers in a multilayer neural network?

- Backpropagation became influential after the 1986 paper by **David Rumelhart, Geoffrey Hinton and Ronald Williams**
 - Demonstrated learning of internal representations

Letter | Published: 09 October 1986

Learning representations by back-propagating errors

[David E. Rumelhart](#), [Geoffrey E. Hinton](#) & [Ronald J. Williams](#)

[Nature](#) **323**, 533–536 (1986) | [Cite this article](#)

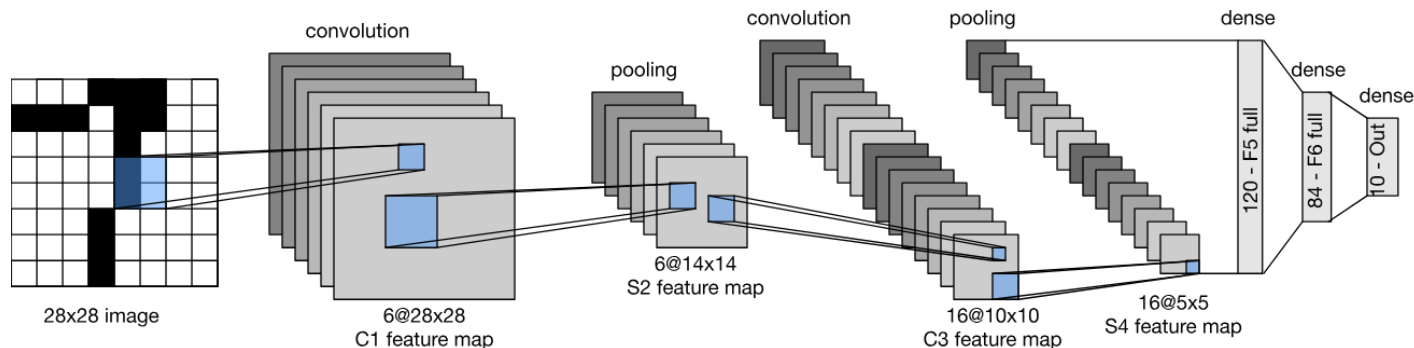
233k Accesses | **24k** Citations | **805** Altmetric | [Metrics](#)

Abstract

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal ‘hidden’ units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure¹.

History (contd.)

- LeNet family (Lecun et al. 1989) represents the **first practical convolutional neural networks (convnets)**.
- Very similar to modern architectures
 - Local receptive fields (convolutions)
 - Weight sharing
 - Hierarchical feature extraction
 - Pooling

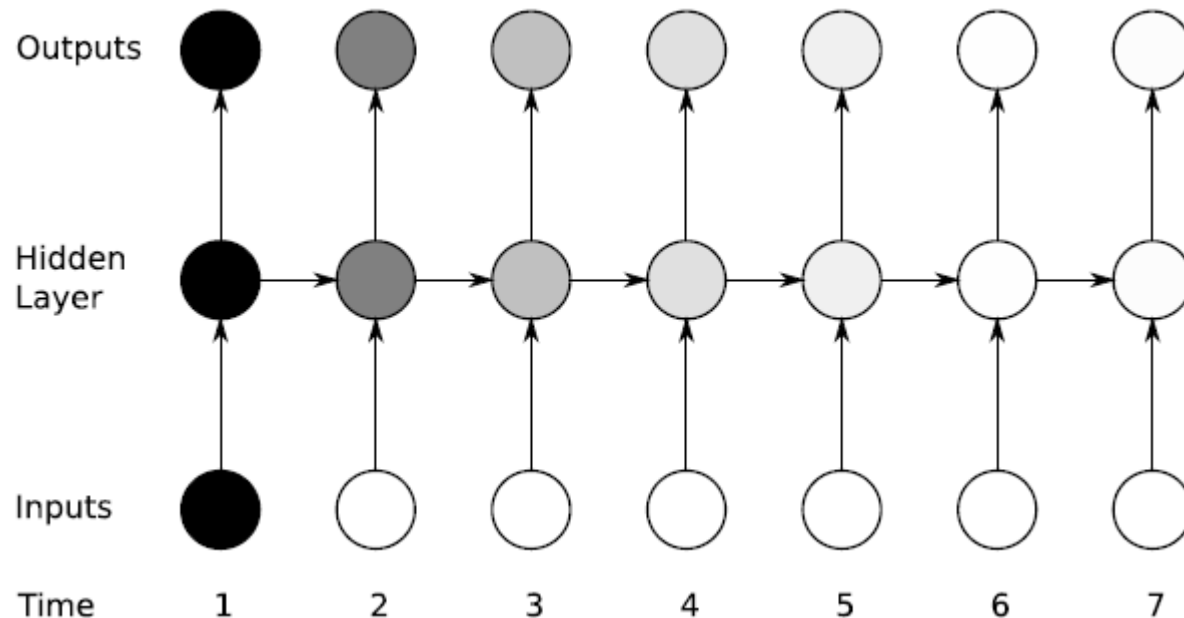


Neural Network Winter

- Computational limits: CNNs were ahead of hardware AND Data requirements
 - To learn good model and representations – needed more layers.
 - To learn more layers – needed more data and compute
 - Data availability was limited – No web, no Youtube, Instagram !
 - CPUs in the 1990s were **orders of magnitude too slow**
 - No GPUs, no CUDA, no distributed training
 - Training even modest CNNs took **weeks**
 - Backpropagation was not robust and fragile – subject to initialization, non convex objective functions, vanishing gradients etc.
- Competing methods worked “well enough” and were theoretically grounded.
 - Support Vector Machines (SVMs), Kernel methods, HMMs for speech
 - Boosting, Random Forests
 - Faster on CPUs and easier to explain

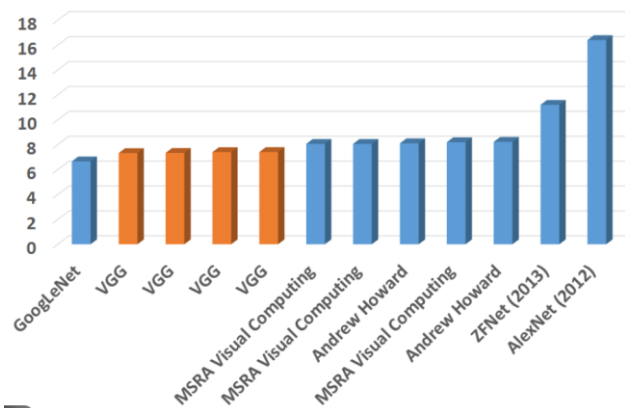
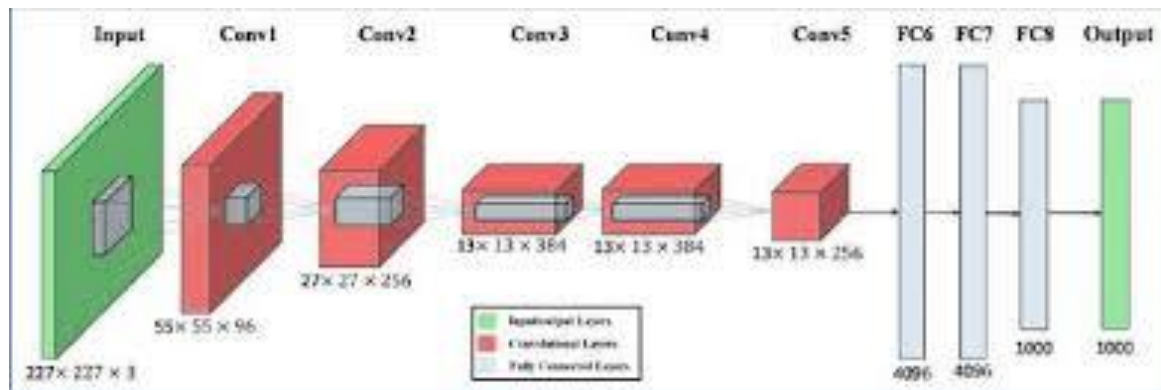
History (contd.)

- Long Short Term Memory (LSTM) (1997) by Hochreiter and Schmidhuber
- This paper directly addressed the **vanishing and exploding gradient problem** in recurrent neural networks.



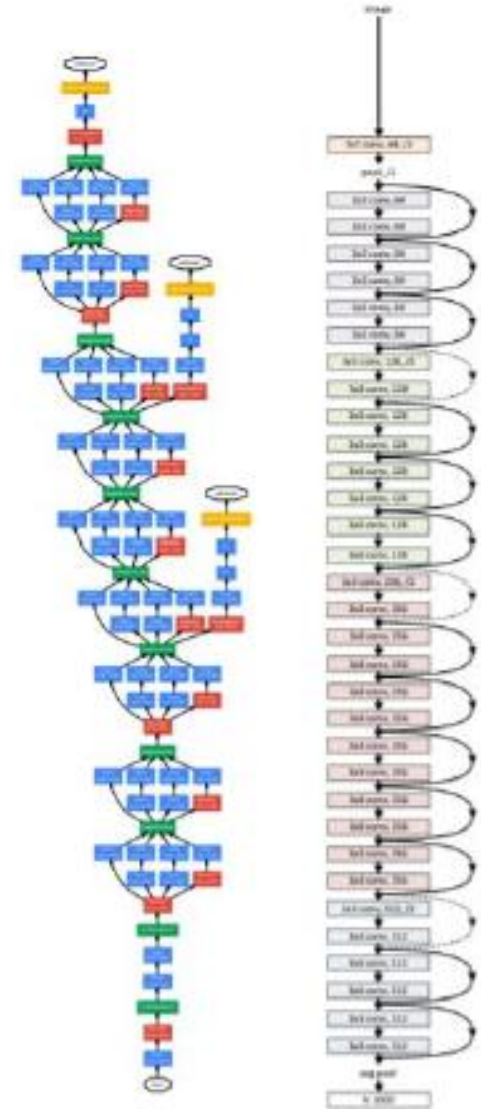
History (contd.)

- AlexNet (2012) by **Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton**
- Network similar to LeNet-5, but of far greater size (80M parameters)
- Implemented using GPUs
- Could beat the SoTA image classification methods by a large margin
- Won **ImageNet Large Scale Visual Recognition Challenge** by a **huge margin**
- First time a neural network **crushed** SIFT/HOG + SVM systems on a flagship benchmark
- **CNNs produce great results when you scale compute, data, and depth together**



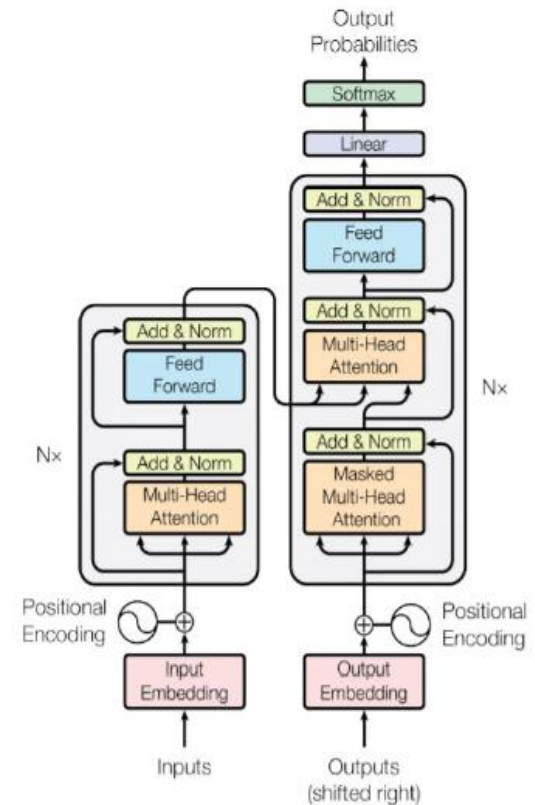
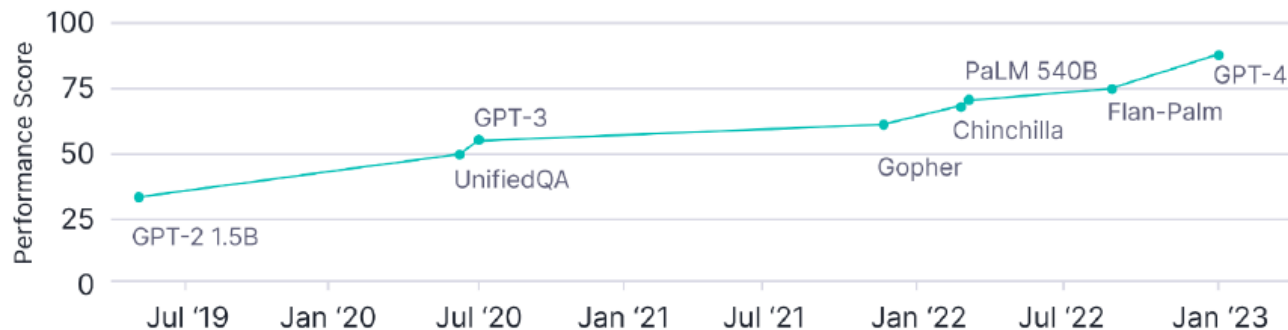
History (contd.)

- AlexNet initiated a trend of more complex and bigger architectures
- GoogLeNet (2015) contains “inception” modules
- ResNet (2015) introduced “skip connections” that facilitate training deeper architectures

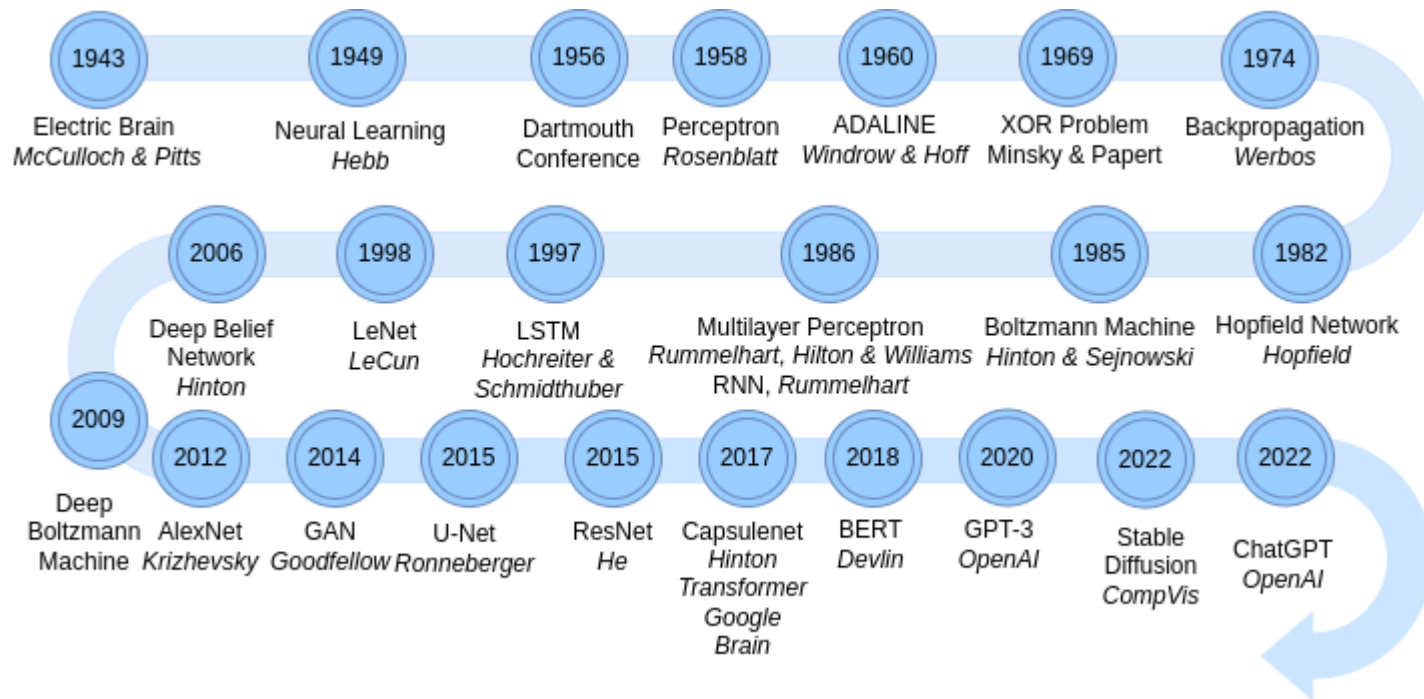


History (contd.)

- Transformers (2017) are attention-based architectures
- Very popular in NLP, and CV
- Some of these models are extremely large - GPT-4 has 1.76 Trillion(estimated), Gemini 1.56 Trillion, PaLM-2 540B parameters



Deep Learning History



Deep Learning - practical perspective

- Doesn't require a deep mathematical grasp (?!)
- Makes the design of large models a system/software development task
- Leverages modern hardware
- Doesn't seem to plateau with more data
- Makes the trained models a commodity